

Data Cleaning

- ❑ Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
- ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ❑ e.g., *Occupation* = " " (missing data)
- ❑ Noisy: containing noise, errors, or outliers
 - ❑ e.g., *Salary* = "-10" (an error)
- ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
 - ❑ *Age* = "42", *Birthday* = "03/07/2010"
 - ❑ Was rating "1, 2, 3", now rating "A, B, C"
 - ❑ discrepancy between duplicate records
- ❑ Intentional (e.g., *disguised missing data*)
 - ❑ Jan. 1 as everyone's birthday?